

Instructions:

You will work on these exercises during the practical sessions in the second week of the course.

Below is an indication of the exercises that we will cover (at least) in each session. We encourage you try them before the session.

- First session (Wednesday/Thursday): exercises 1, 2.
- Second session (Friday): exercises 4, 5.

Exercise 1: MAP Solution for Linear Regression

In this exercise, you will solve and practice with the maximum a posteriori (MAP) estimator for linear regression with basis functions. We assume N training vectors $\{\mathbf{x}_n\}_{n=1}^N$, each mapped to a feature vector

$$\phi_n = (\phi_0(\mathbf{x}_n), \phi_1(\mathbf{x}_n), \dots, \phi_{M-1}(\mathbf{x}_n))^T,$$

using basis functions $\phi_j(\mathbf{x})$ for $j = 0, \dots, M-1$, where $\phi_0(\mathbf{x}) = 1$ is a bias basis. The training set consists of input-output pairs $(\mathbf{x}_n, t_n)$. The regression prediction is

$$y(\mathbf{x}_n, \mathbf{w}) = \mathbf{w}^T \phi_n,$$

and the data samples are independent.

The likelihood is Gaussian,

$$p(\mathbf{t} \mid \Phi, \mathbf{w}, \beta) = \mathcal{N}(\mathbf{t} \mid \Phi \mathbf{w}, \beta^{-1} \mathbf{I}),$$

and so is the prior over $\mathbf{w}$:

$$p(\mathbf{w}) = \mathcal{N}(\mathbf{w} \mid \mathbf{0}, \alpha^{-1} \mathbf{I}).$$

The MAP solution for the weights is

$$\mathbf{w}_{\text{MAP}} = (\Phi^T \Phi + \lambda \mathbf{I})^{-1} \Phi^T \mathbf{t},$$

with

$$\Phi = \begin{pmatrix} \phi_0(\mathbf{x}_1) & \phi_1(\mathbf{x}_1) & \cdots & \phi_{M-1}(\mathbf{x}_1) \\ \vdots & \vdots & \ddots & \vdots \\ \phi_0(\mathbf{x}_N) & \phi_1(\mathbf{x}_N) & \cdots & \phi_{M-1}(\mathbf{x}_N) \end{pmatrix}.$$

Derive the MAP solution by answering the following questions.

- (a) Write down the likelihood $p(\mathcal{D} \mid \mathbf{w})$ using: (a) a product over N , and (b) vector/matrix form. You may start with the product form and simplify to the matrix form. Define any matrices and vectors you use.
- (b) Write down the explicit form of the prior $p(\mathbf{w})$ using the multivariate Gaussian density with the correct mean and covariance. Compute $\ln p(\mathbf{w})$.
- (c) Write down an expression for the posterior $p(\mathbf{w} \mid \mathcal{D})$ by applying Bayes' rule. You do not need to write out the explicit Gaussian densities. Show that the evidence requires an integral, which you do not need to solve analytically, and then replace it with the corresponding probability distribution.
- (d) Compute the log-posterior for both likelihood expressions from the first question. Collect all terms independent of $\mathbf{w}$ into a constant C . Which parts do not depend on $\mathbf{w}$? Why is finding the MAP simpler than finding the full posterior distribution?

- (e) Solve for $\mathbf{w}_{\text{MAP}}$ by taking the derivative of the log-posterior with respect to $\mathbf{w}$, setting it to zero, and solving for $\mathbf{w}$. Do this for both forms of the log-posterior.
- (f) The prior for $\mathbf{w}$ assumes the same distribution for each entry, including w_0 , which is multiplied by the basis function $\phi_0(\mathbf{x}) = 1$. What is the role of w_0 and $\phi_0(\mathbf{x})$? Why should we avoid placing the same penalty/prior on this basis? Rewrite $p(\mathbf{w})$ so that w_0 has its own prior/penalty.

Exercise 2: Absolute Error for linear regression

Recall the least-squares error function for linear regression:

$$E(\mathbf{w}) = \frac{1}{2}(y - \mathbf{w}^\top \mathbf{x})^2.$$

This is the error associated with a single training sample with input $\mathbf{x}$ and target value y .

An alternative is the absolute error:

$$J(\mathbf{w}) = |y - \mathbf{w}^\top \mathbf{x}|.$$

- (a) Which of the two error functions is more robust to outliers? Why?
- (b) Derive the gradient-descent update rule for this new objective function. You can assume $y - \mathbf{w}^\top \mathbf{x} \neq 0$.
- (c) What happens when $y - \mathbf{w}^\top \mathbf{x} = 0$? What is a reasonable way to define the update rule in this case?

Exercise 3: Bayesian Linear Regression with Laplace Prior

Exam question

Consider the linear regression model

$$\mathbf{y} = \mathbf{X}\mathbf{w} + \boldsymbol{\epsilon}, \quad \boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \sigma^2 \mathbf{I}_N),$$

where $\mathbf{X} \in \mathbb{R}^{N \times D}$ is a data matrix consisting of N i.i.d. samples, $\mathbf{y} \in \mathbb{R}^N$ is the vector of targets, and $\mathbf{w} \in \mathbb{R}^D$ is the vector of coefficients. Each coefficient w_d , for $d \in \{1, \dots, D\}$, has an independent Laplace prior

$$p(w_d) = \frac{\alpha}{2} \exp(-\alpha |w_d|), \quad \alpha > 0.$$

Show that the optimization problem for the MAP estimator is

$$\hat{\mathbf{w}}_{\text{MAP}} = \arg \min_{\mathbf{w} \in \mathbb{R}^D} \frac{1}{2\sigma^2} \|\mathbf{y} - \mathbf{X}\mathbf{w}\|_2^2 + \alpha \|\mathbf{w}\|_1.$$

Exercise 4: Posterior Uncertainty in Bayesian Linear Regression

We consider the Bayesian linear regression model presented in the lectures and in Section 3.3. of Bishop [1], where the posterior precision matrix after observing N data points is

$$\mathbf{S}_N^{-1} = \alpha \mathbf{I} + \beta \sum_{n=1}^N \phi(\mathbf{x}_n) \phi(\mathbf{x}_n)^\top,$$

where $\alpha, \beta > 0$ are the prior and observation precisions,¹ respectively. The variance of the predictive distribution for the target at a new input $\mathbf{x}$ is

$$\sigma_N^2(\mathbf{x}) = \frac{1}{\beta} + \phi(\mathbf{x})^\top \mathbf{S}_N \phi(\mathbf{x}).$$

The term $1/\beta$ represents the noise on the data, while $\phi(\mathbf{x})^\top \mathbf{S}_N \phi(\mathbf{x})$ is the uncertainty associated with the model weights.

- (a) Show that $\mathbf{S}_N$ is positive definite.
- (b) Show that adding one more training point $\mathbf{x}_{N+1}$ cannot increase the uncertainty of the predictive distribution, i.e.,

$$\sigma_{N+1}^2(\mathbf{x}) \leq \sigma_N^2(\mathbf{x}).$$

Hint: Use the following matrix identity: $(\mathbf{M} + \mathbf{v}\mathbf{v}^\top)^{-1} = \mathbf{M}^{-1} - \frac{(\mathbf{M}^{-1}\mathbf{v})(\mathbf{v}^\top\mathbf{M}^{-1})}{1 + \mathbf{v}^\top\mathbf{M}^{-1}\mathbf{v}}$

Adapted from Chapter 3 of Pattern Recognition and Machine Learning [1].

Exercise 5: Bias and Variance

Explain what effect the following operations will have on the bias and variance of your model. Fill in one of “increases”, “decreases”, or “no change” in each of the cells.

Operation	Bias	Variance	Justification
Regularizing the weights in a linear/logistic regression model			
Pruning a decision tree (to a certain depth, for example)			
Using dropout to train a deep neural network			
Increasing k in k -nearest neighbor models			
Averaging multiple decision trees in a random forest			
Increasing the degree of polynomial regression			
Increasing the number of hidden units in a neural network			
Using fewer training examples			

References

- [1] Christopher M. Bishop. *Pattern Recognition and Machine Learning*. Springer, New York, 2006.

¹The precision is the inverse of the variance.